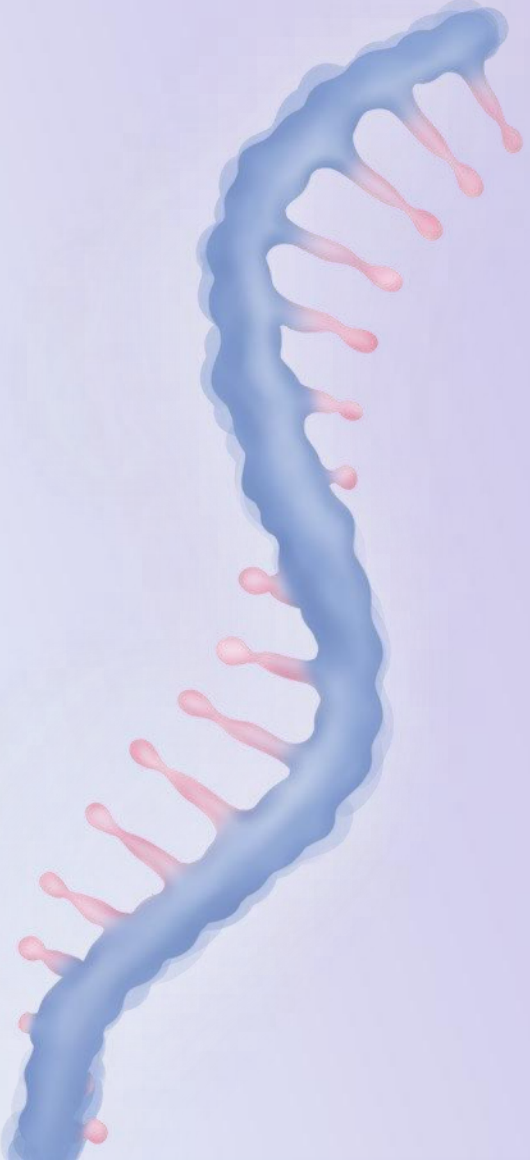




Designing Improved Mechanisms for Structurally Interacting RNAs (sxRNAs)

Thesis Talk
May 19 2025

Saad Waheed '25
Professor Daniel Aalberts

RNA-based Therapeutics

- **RNA-based therapeutics** act as blueprints, directing cells to produce specific proteins that function as drugs.
- Therapeutic benefit often requires expression only in the intended cell type, such as cancer cells in the lung or liver. **Expression in non-target regions can have adverse side effects.**

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As seen in...

Example: Covid Vaccine

The primary target for mRNA-based COVID vaccines are specific immune cells, called dendritic cells, which cannot be specifically targeted. Much of the Spike protein is therefore expressed in muscle and connective tissue, reducing effectiveness.

The muscle soreness many experienced from the Covid vaccine results from the Spike protein being produced in muscle tissue cells.

Muscles



Effect: Soreness

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As seen in...

Leads to a need for

Cell Specificity

- **Existing drug delivery systems lack precision**, dispersing RNA systemically, that leads to off-target uptake and side effects.
- Alternative work-around: achieve cell-specific expression of RNA-based drugs.

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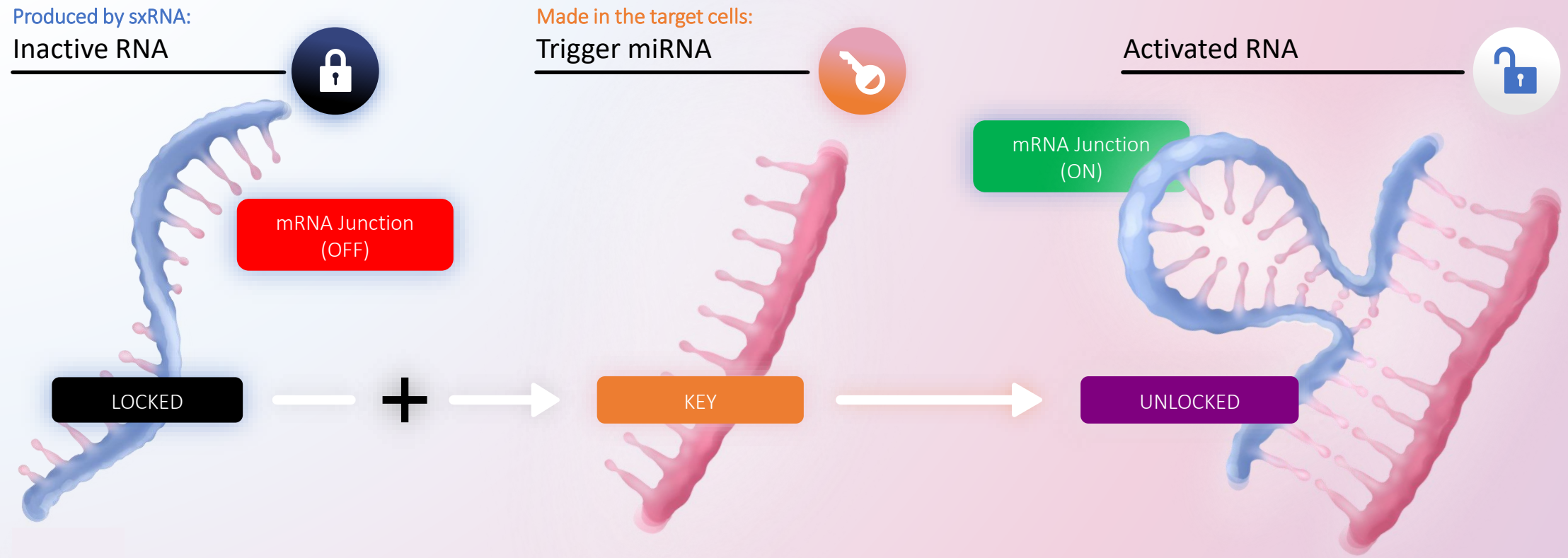
Muscles



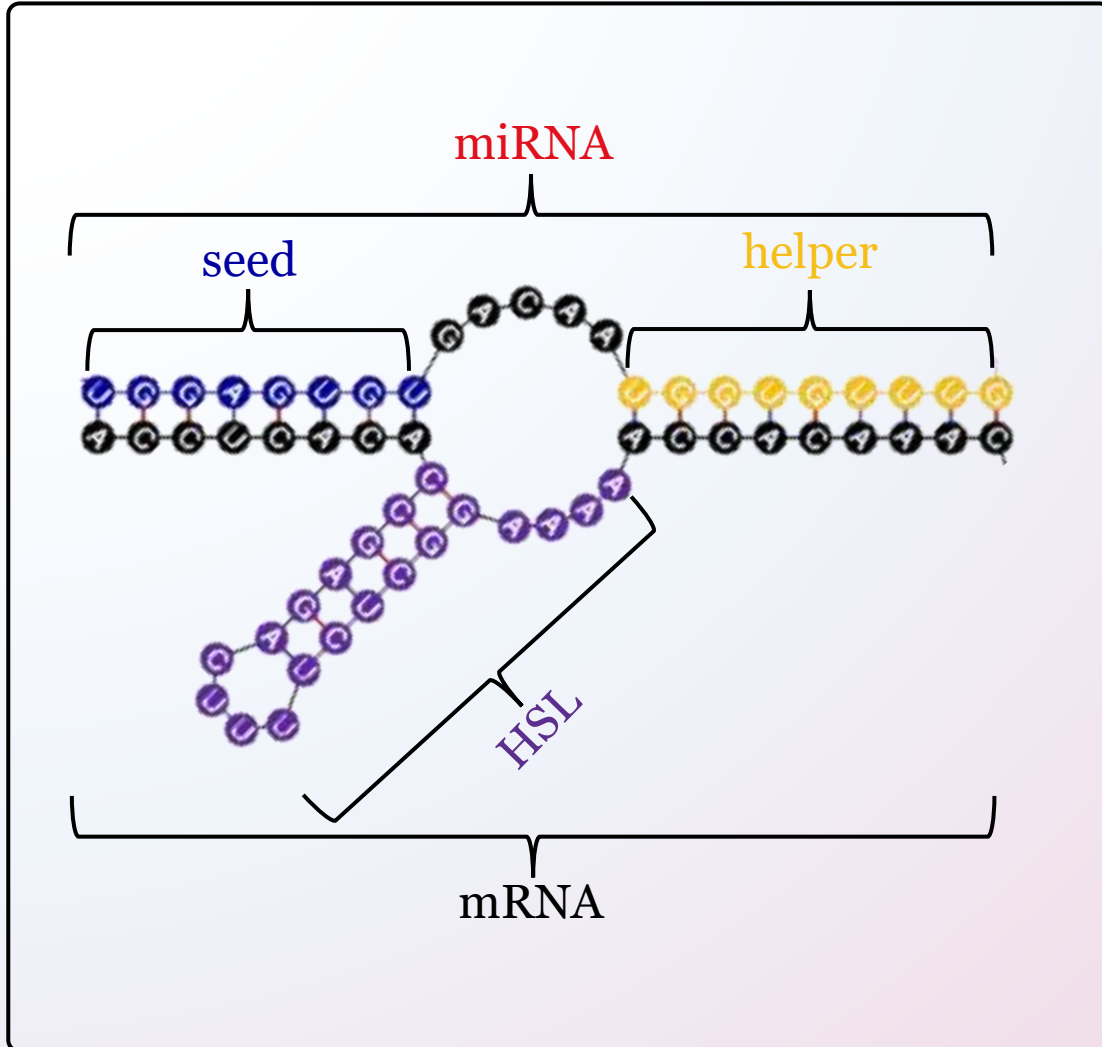
Effect: Soreness

sxRNA – a Key and Lock Mechanism Proposed in Doyle *et al.* 2017

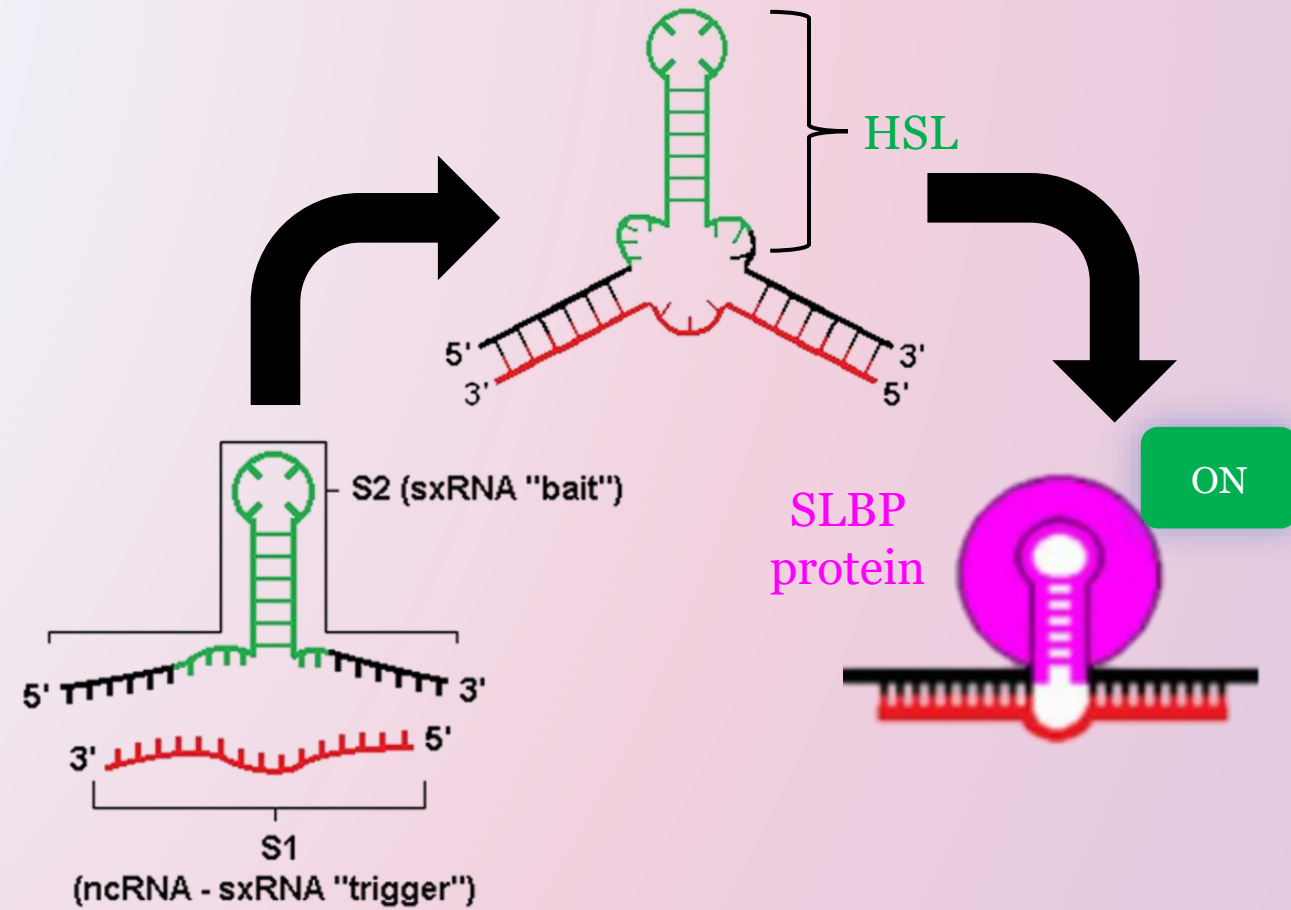
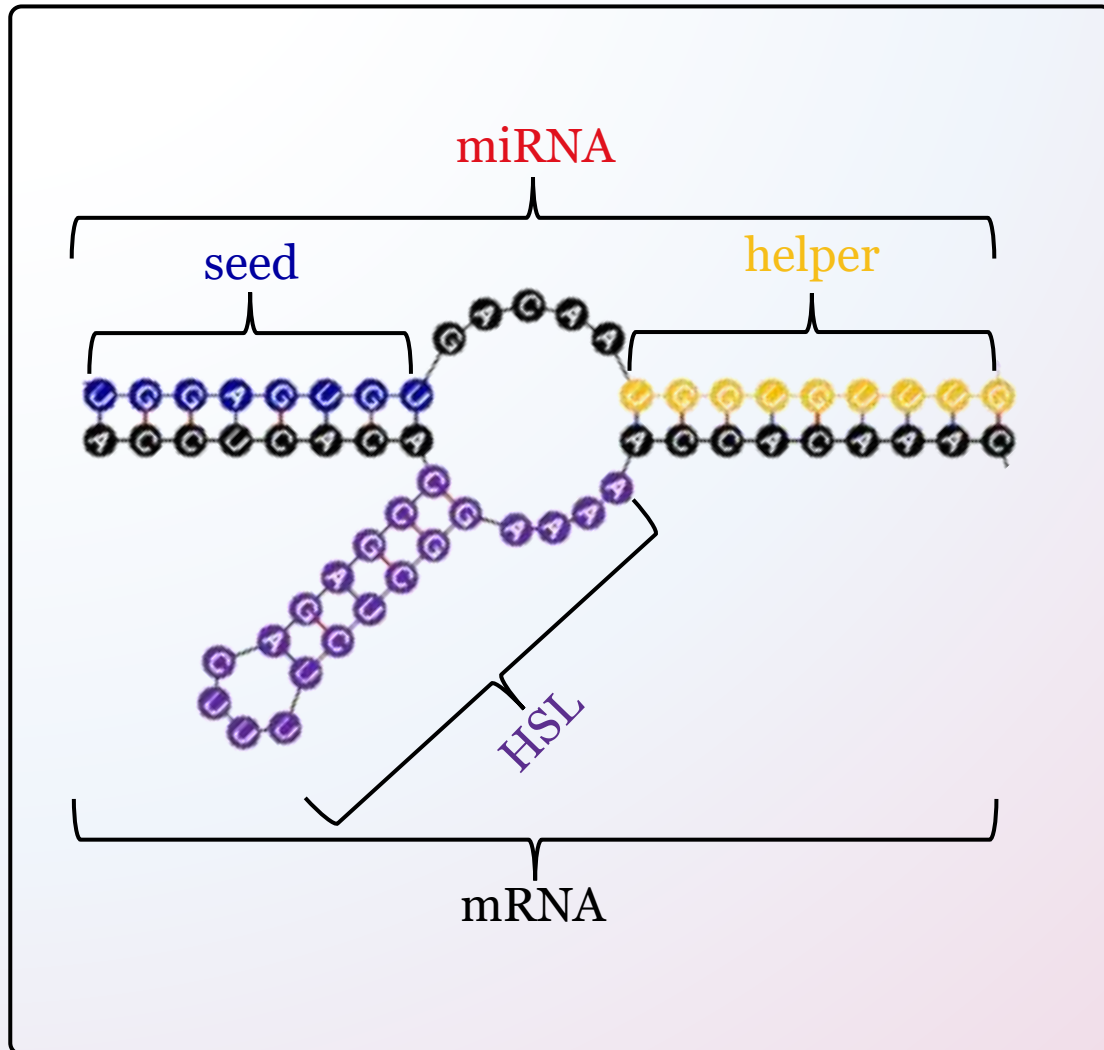
- Developed by our experimental partners, **sxRNA Technologies Inc.**, to LOCK and UNLOCK RNA therapeutics for cell-specific activity.



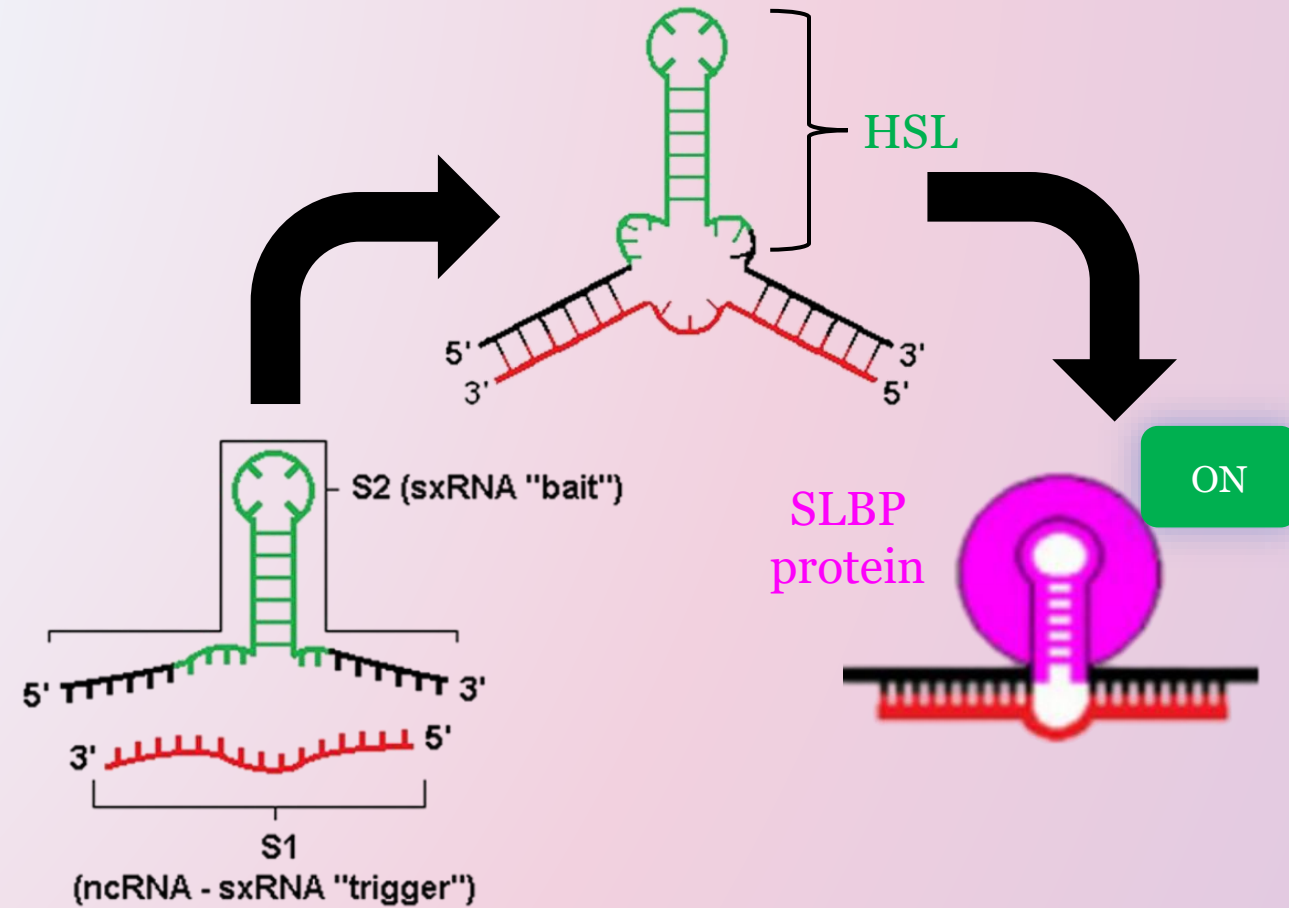
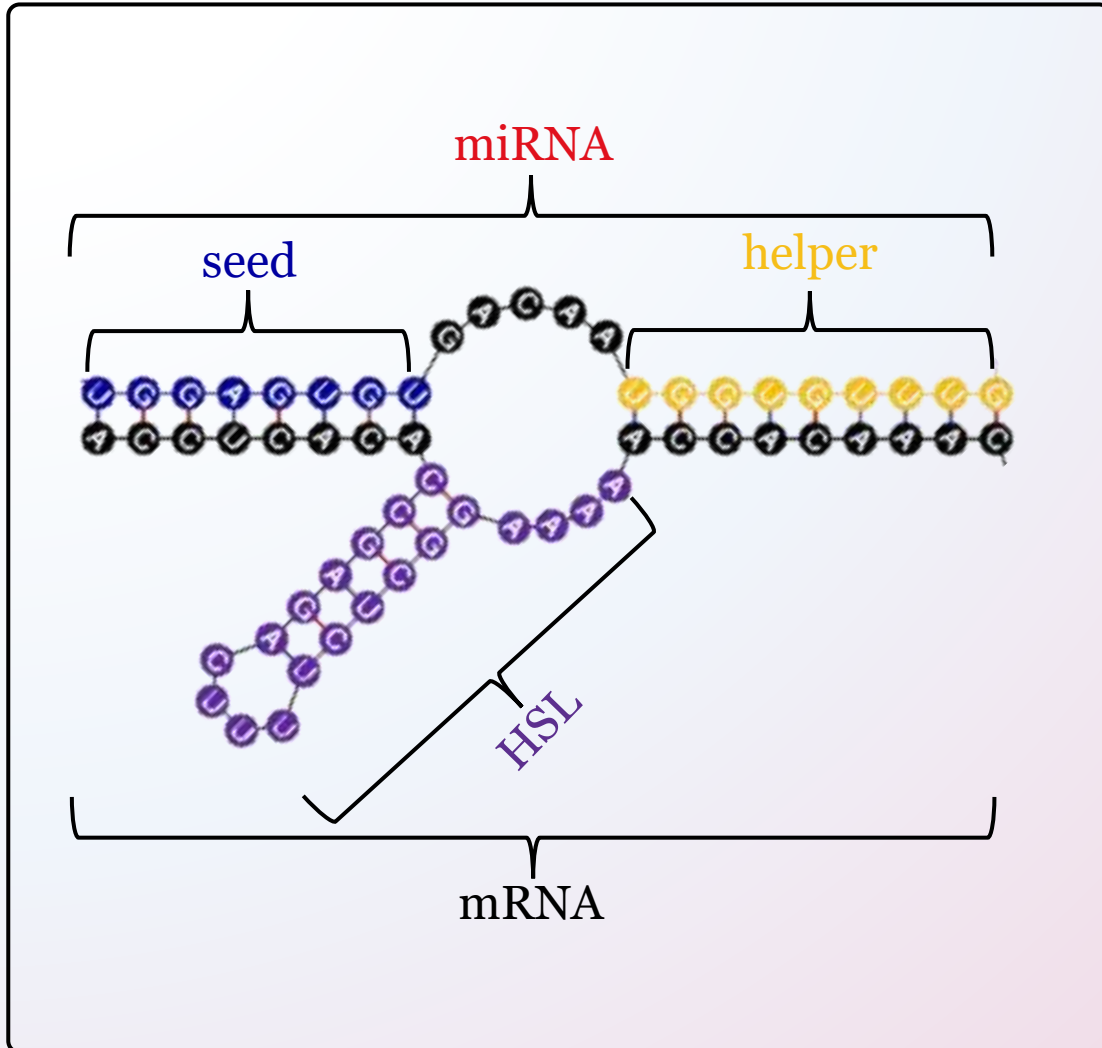
A Closer Look at the 3 Way Junction (3WJ) mechanism



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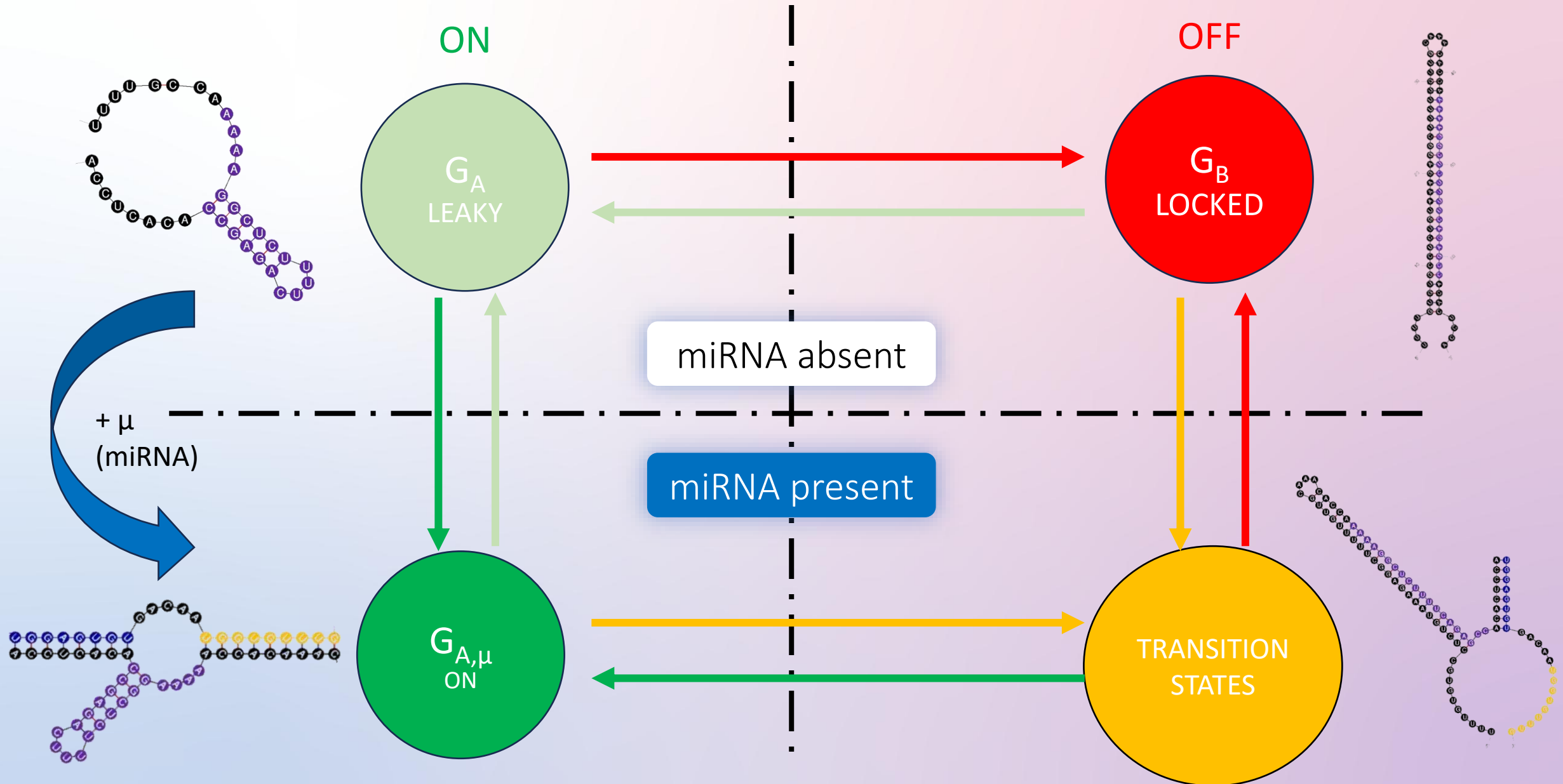


A Closer Look at the 3 Way Junction (3WJ) mechanism

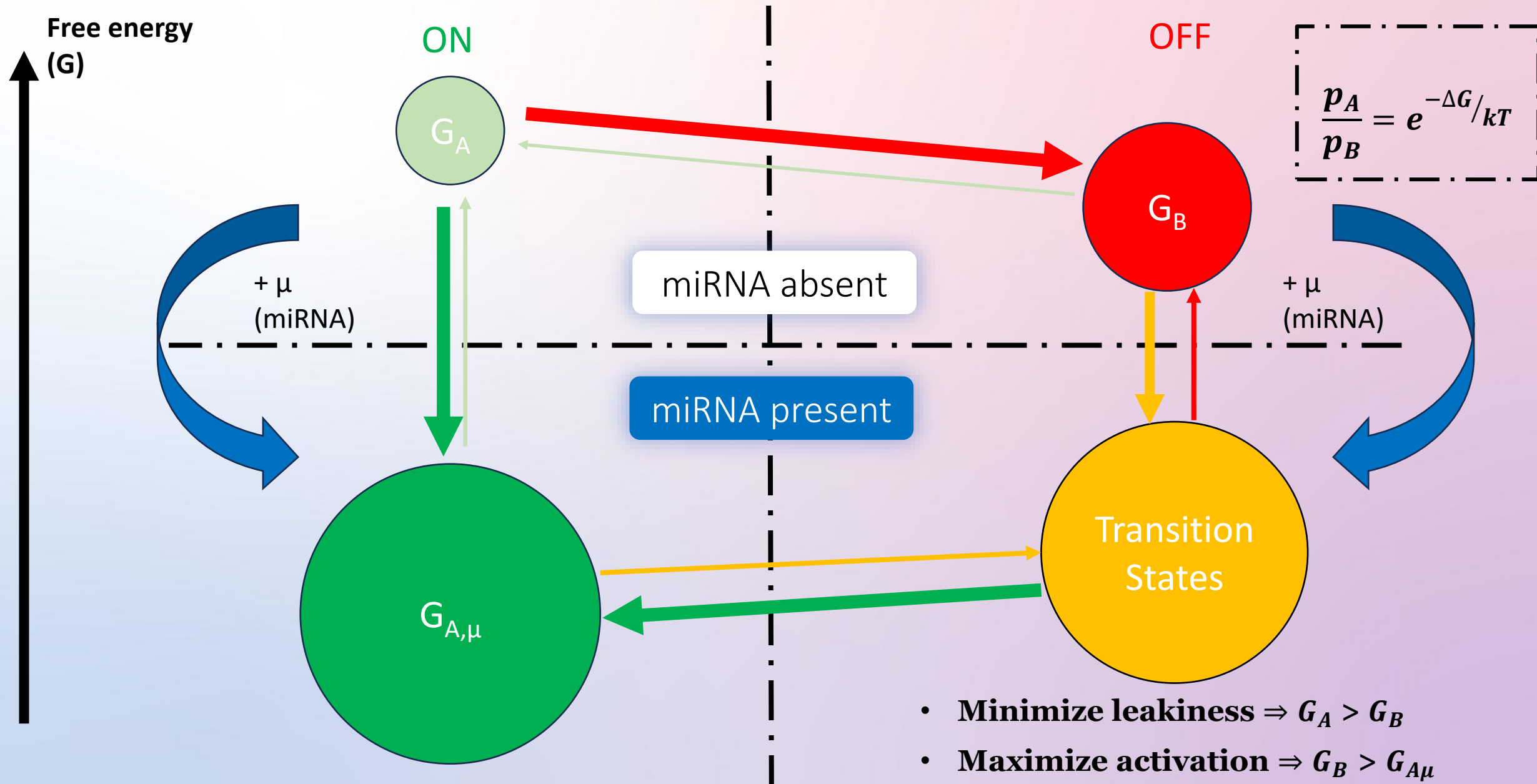


- **Issue:** Treats “lock” and “key” as definite states, not taking into account thermal fluctuations.
- Experimental data shows non-significant activation even without miRNA.

Competing States in Both OFF and ON Cases

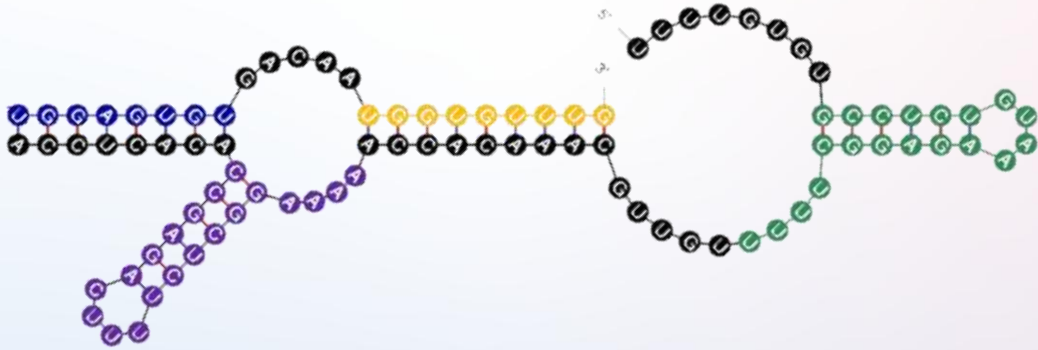


The Free Energy Landscape We Want

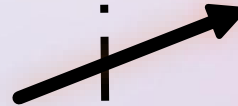


Base Pair Representation Through Bow Diagrams

$Z_{A\mu}$: ON



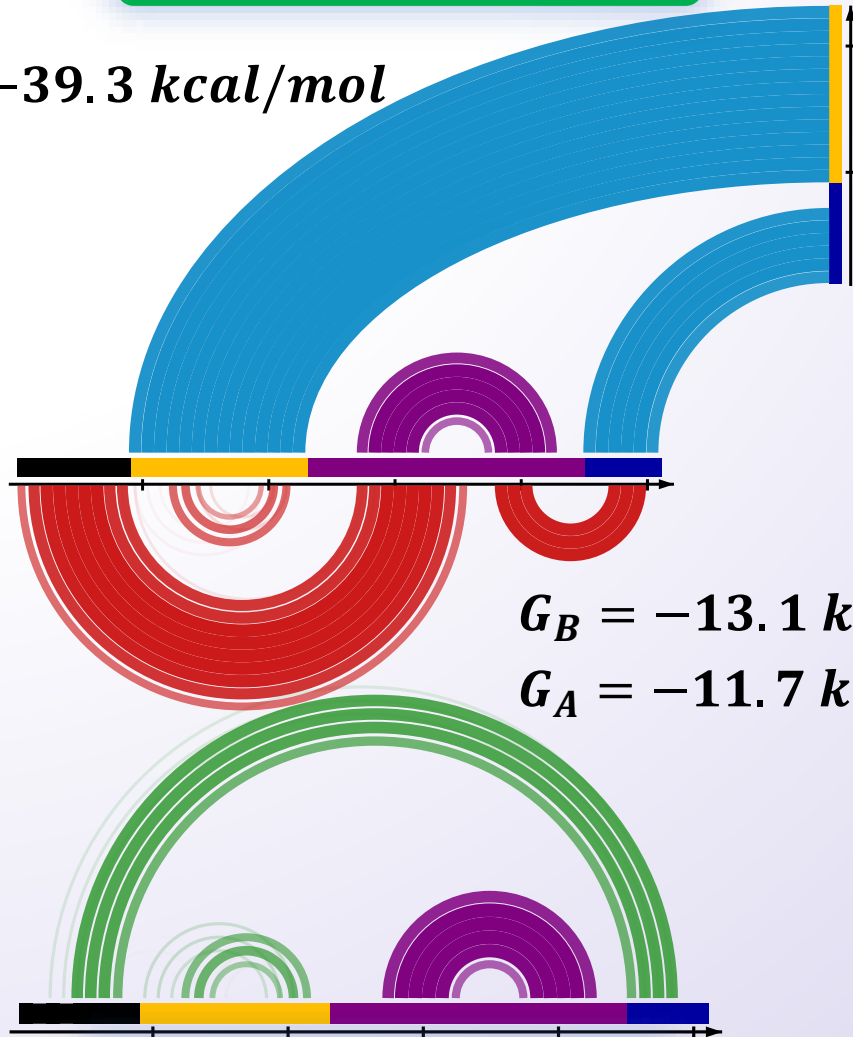
Z_B : OFF



Applying Theory to Results from Doyle *et al.* 2017

BR-1e: Least Leaky

$$G_{A\mu} = -39.3 \text{ kcal/mol}$$



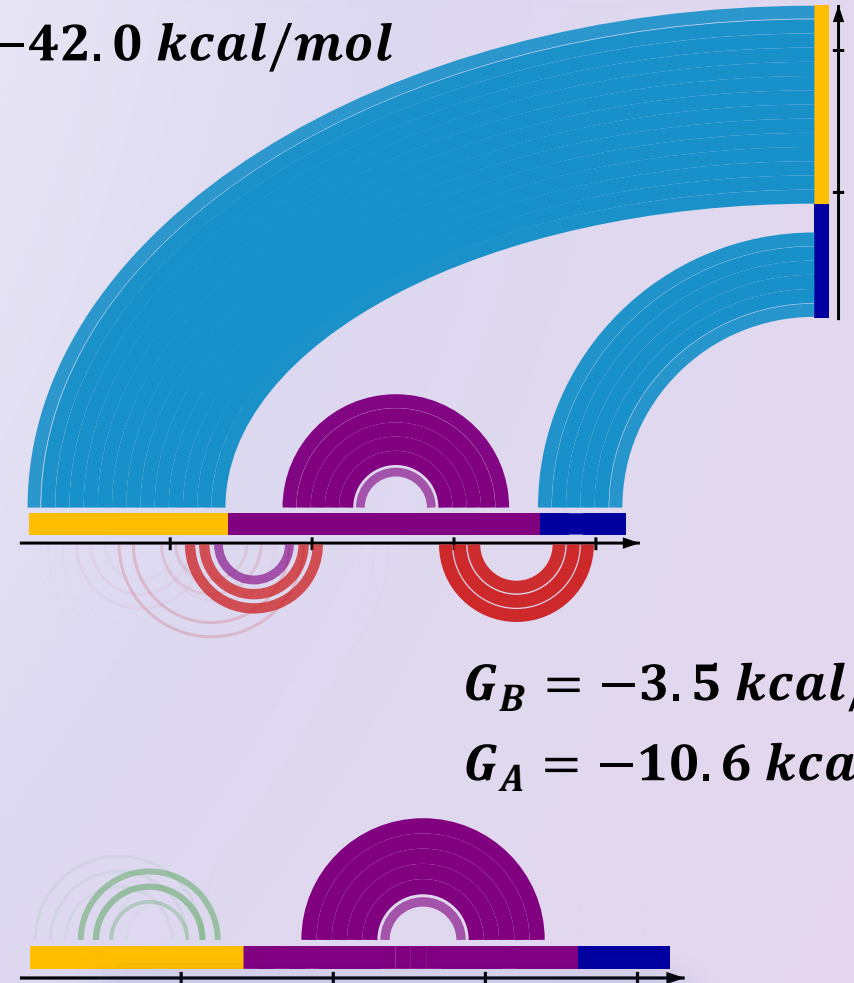
$$G_B = -13.1 \text{ kcal/mol}$$

$$G_A = -11.7 \text{ kcal/mol}$$

$$G_A > G_B > G_{A\mu}$$

BR-1a: Most Leaky

$$G_{A\mu} = -42.0 \text{ kcal/mol}$$

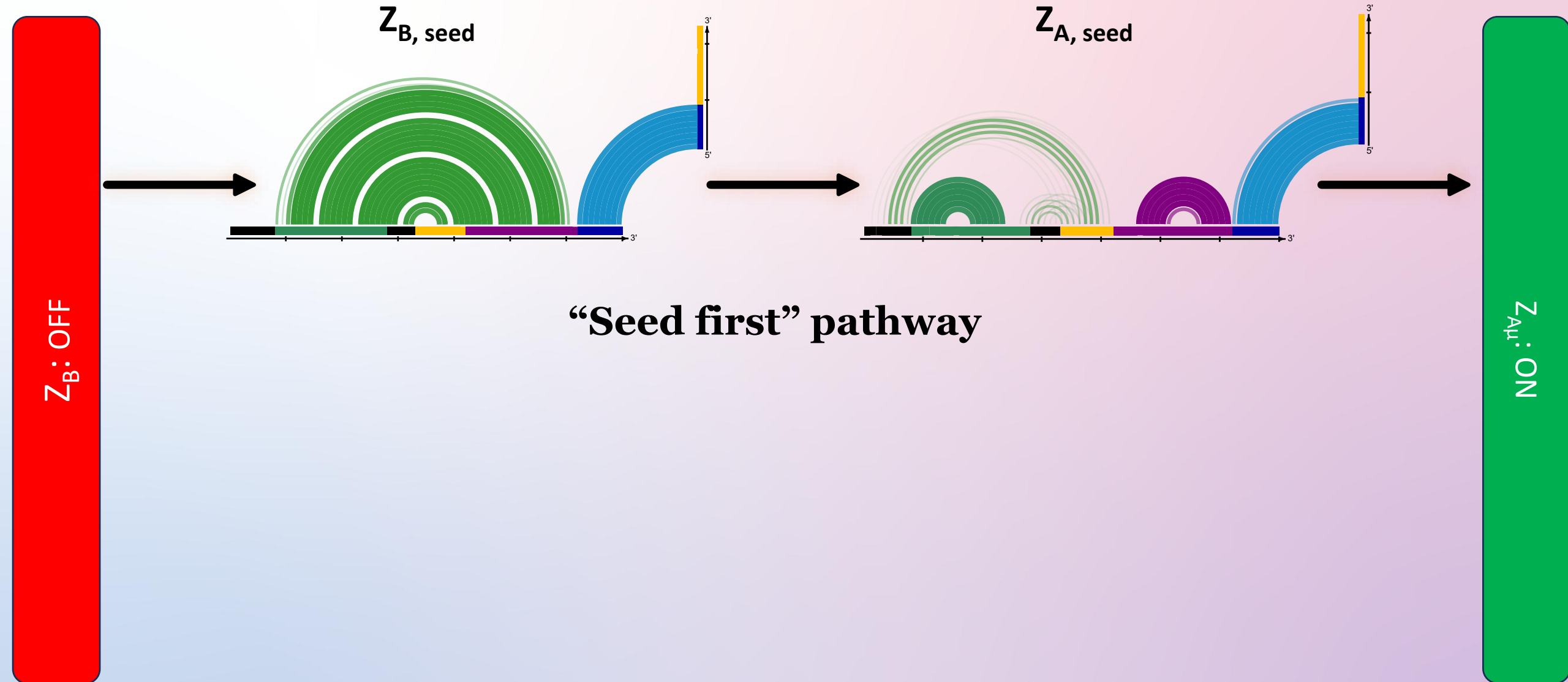


$$G_B = -3.5 \text{ kcal/mol}$$

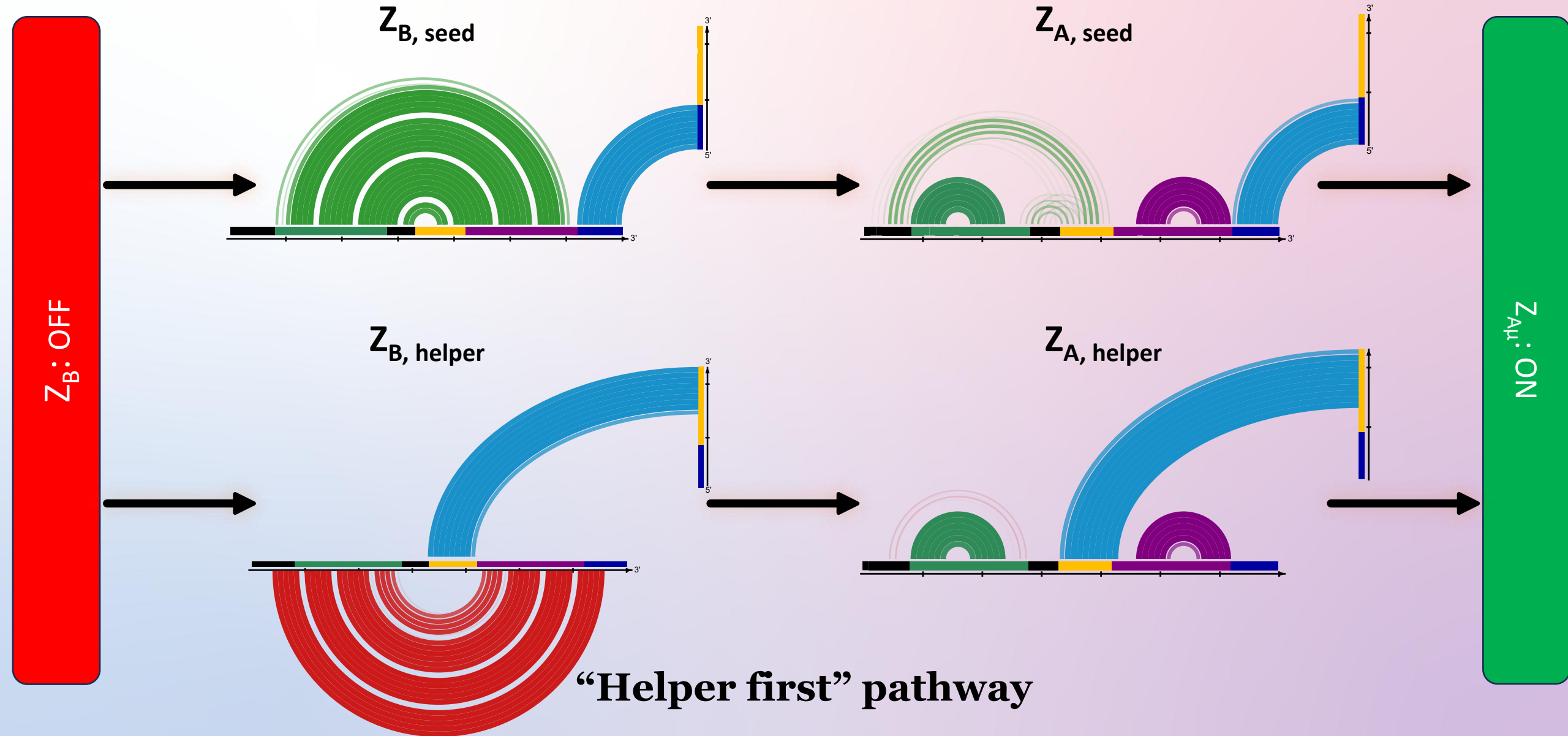
$$G_A = -10.6 \text{ kcal/mol}$$

$$G_B > G_A > G_{A\mu}$$

Transition Pathways from OFF to ON

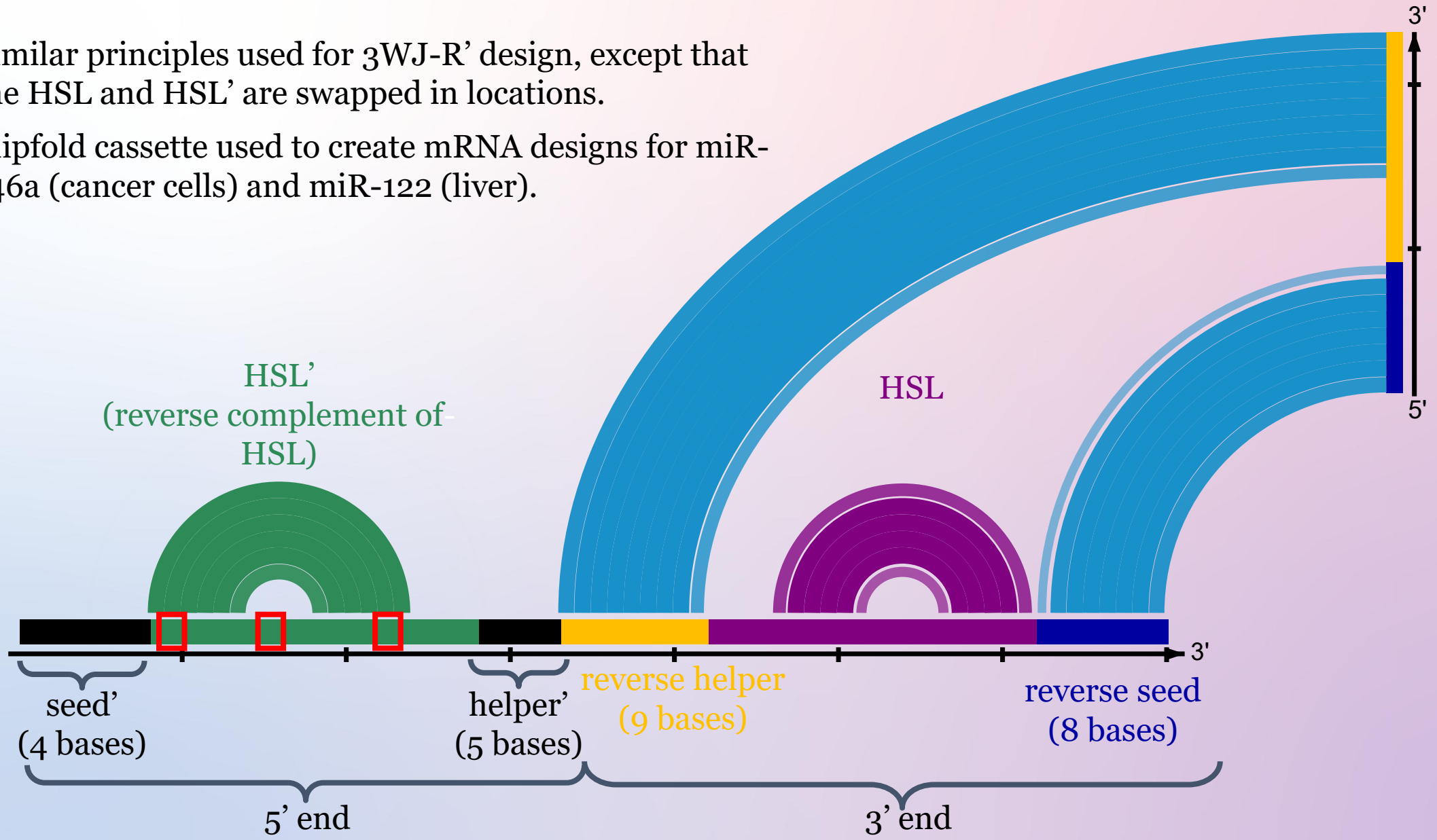


Transition Pathways from OFF to ON



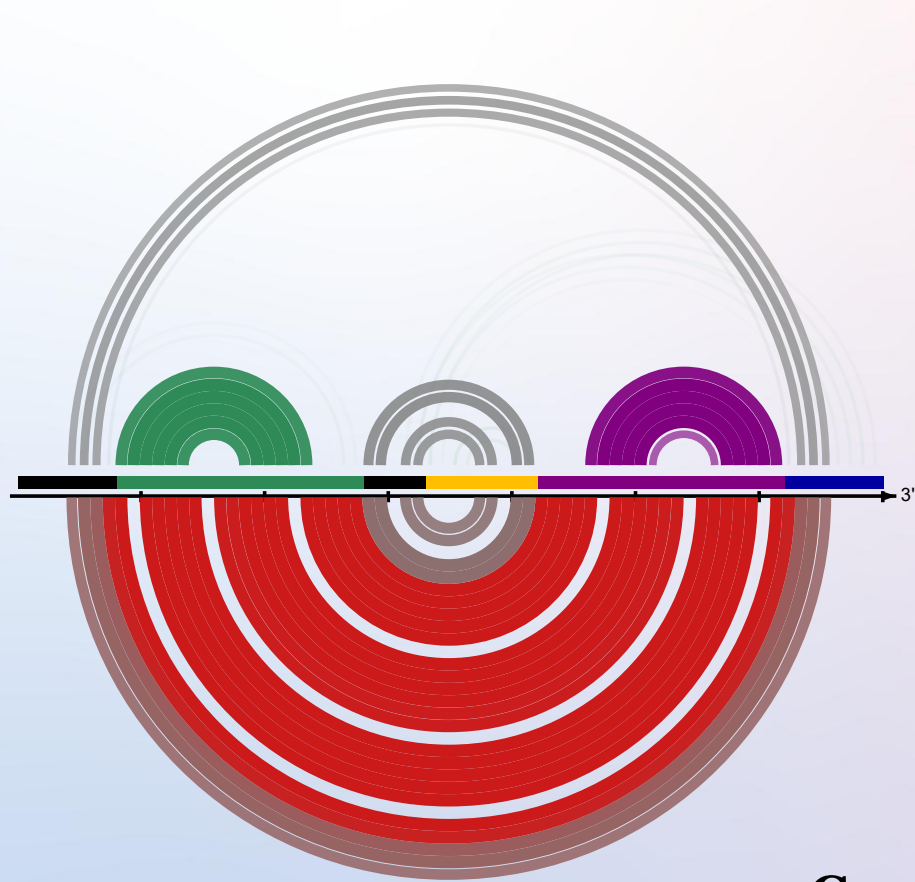
The Flipfold 3WJ Architecture

- Similar principles used for 3WJ-R' design, except that the HSL and HSL' are swapped in locations.
- Flipfold cassette used to create mRNA designs for miR-146a (cancer cells) and miR-122 (liver).

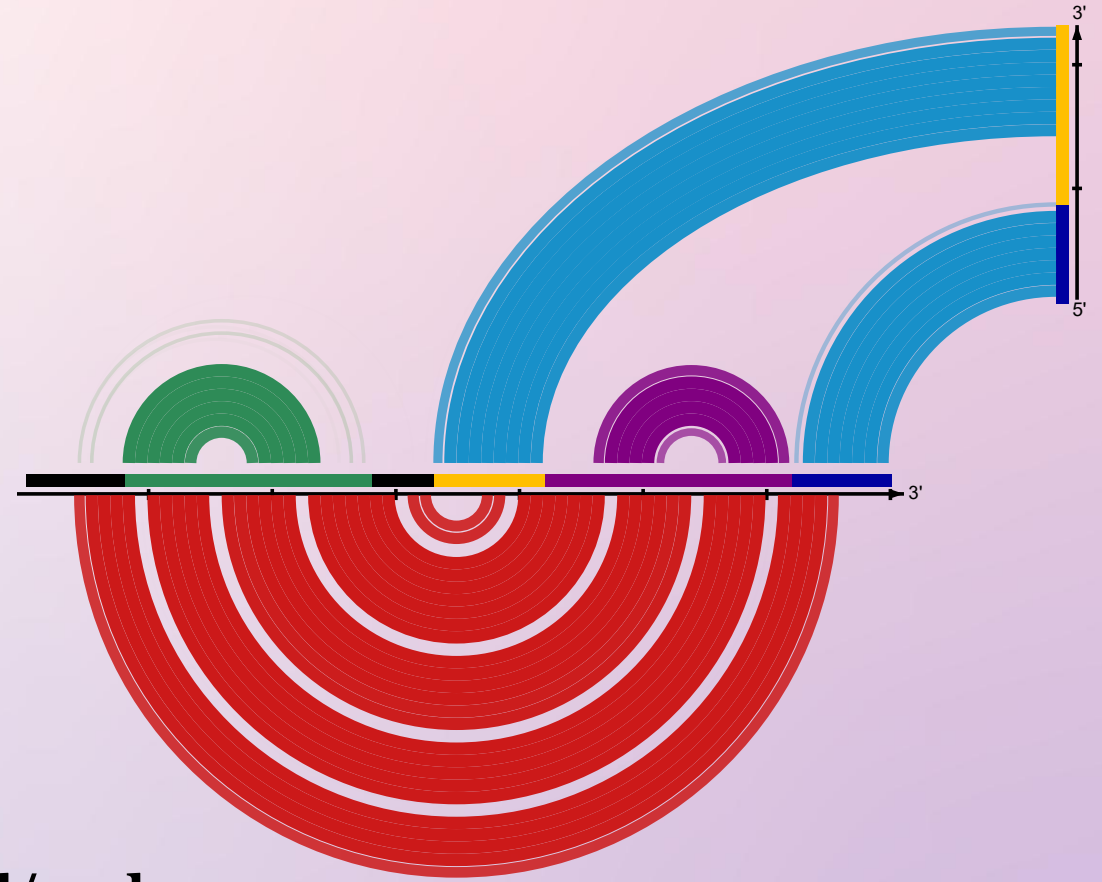


Predictions for miR-146

$$G_A = -23.77 \text{ kcal/mol}$$



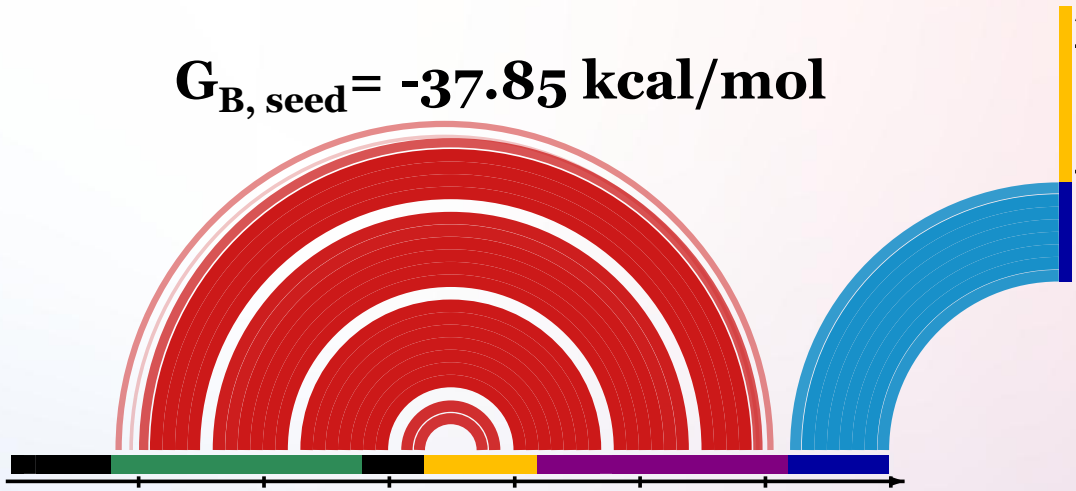
$$G_{A\mu} = -46.94 \text{ kcal/mol}$$



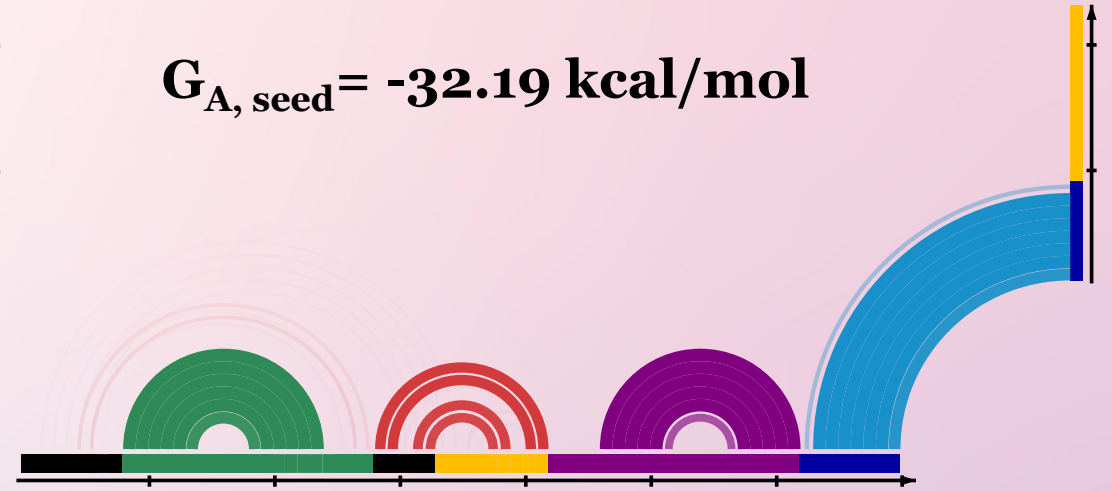
$$G_B = -33.15 \text{ kcal/mol}$$

Predictions for miR-146 (continued)

$G_{B, \text{seed}} = -37.85 \text{ kcal/mol}$

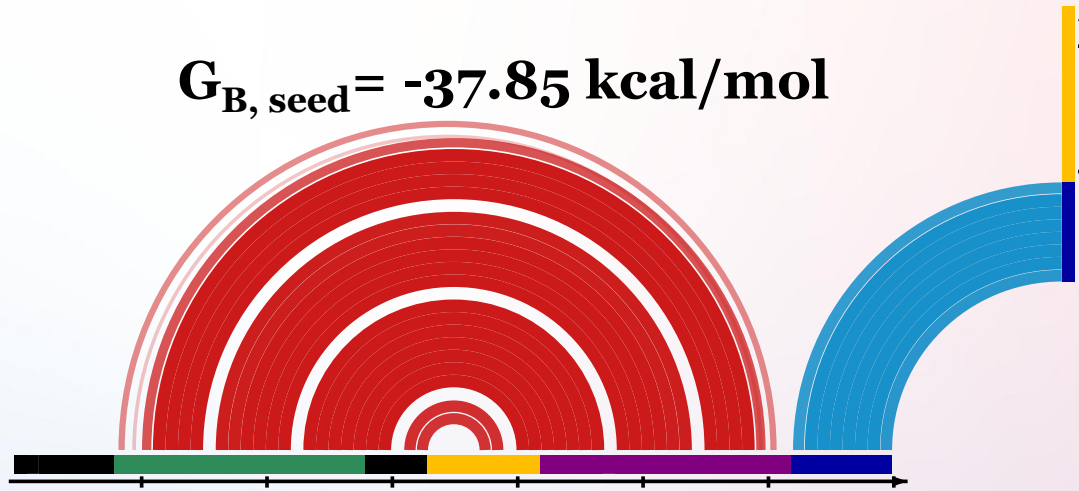


$G_{A, \text{seed}} = -32.19 \text{ kcal/mol}$

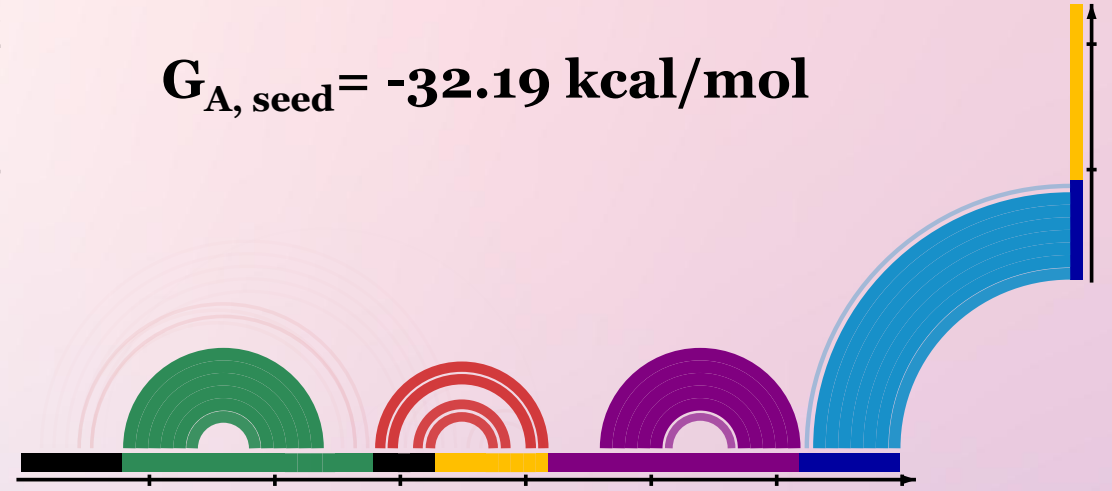


Predictions for miR-146 (continued)

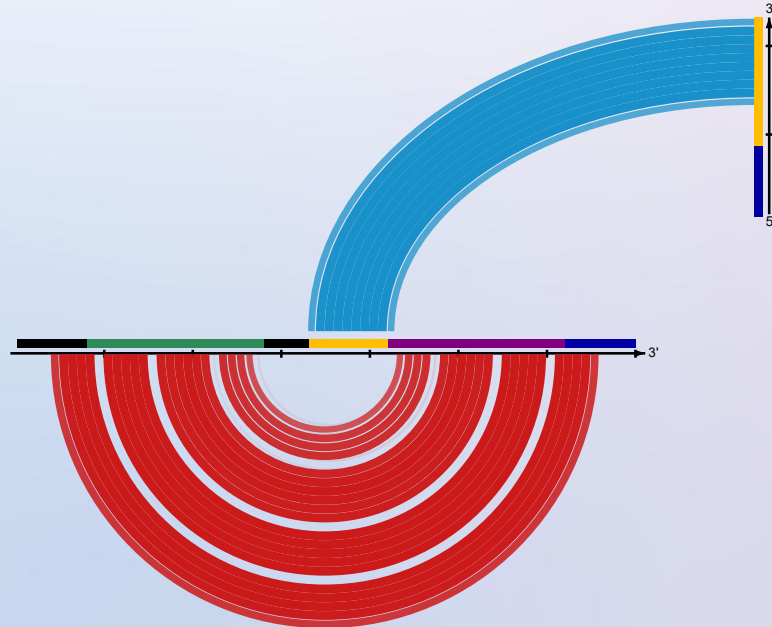
$G_{B, \text{seed}} = -37.85 \text{ kcal/mol}$



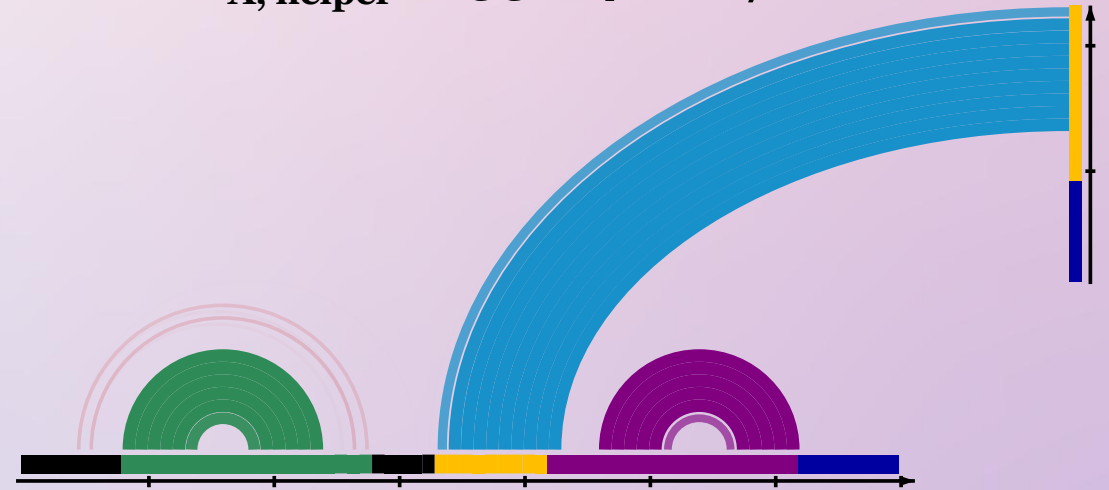
$G_{A, \text{seed}} = -32.19 \text{ kcal/mol}$



$G_{B, \text{helper}} = -36.47 \text{ kcal/mol}$



$G_{A, \text{helper}} = -33.84 \text{ kcal/mol}$



Acknowledgements

- This research is being conducted in collaboration with **sxRNA Technologies Inc.** and **Professor Scott Tenenbaum**'s research group at the University of Albany.
- I would like to greatly thank **Professor Daniel Aalberts** for his invaluable guidance throughout the semester, and **Professor Catherine Kealhofer** for her patience as my second reader.

Citations

- **Doyle et al. (2017)**
Francis Doyle, Sameer Lapsia, Salvatore Spadaro, Zachary E. Wurz, Sumita Bhaduri-McIntosh & Scott A. Tenenbaum. *Engineering Structurally Interacting RNA (sxRNA)*. **Scientific Reports** 7, 45393. doi:10.1038/srep45393
- **Marzluff, Wagner & Duronio (2008)**
William F. Marzluff, Eric J. Wagner & Robert J. Duronio. *Metabolism and regulation of canonical histone mRNAs: life without a poly(A) tail*. **Nature Reviews Genetics** 9, 843–854. doi:10.1038/nrg2438
- **Markham & Zuker (2008)**
M. Markham & M. Zuker. *UNAFold: software for nucleic acid folding and hybridization*. **Methods in Molecular Biology** 453, 3–31.